

Hana Fujimoto

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Machine Learning Engineer (formerly civil engineering)

Skills

python, pytorch, onnx, opencv, cuda, numpy, ray, mlflow, polars

Experience

Role	Organisation	Location	Period	Detail
Machine Learning Engineer	Sakura Vision	Osaka, Japan	Apr 2023 – Present	Ships defect-detection models to 40 factory edge boxes across seven client sites in Kansai and Chubu. Cut inference cost 60% by exporting to ONNX and quantising to INT8, with a calibration set per line. Owns the retraining loop: Ray for sweeps, MLflow for the registry, and a shadow-deploy gate before promotion. Rewrote the data layer on Polars; nightly aggregation went from 50 minutes to under four. Wrote the drift monitor that catches lens fouling before the false-positive rate reaches the operators.
Computer Vision Engineer	Rinku Robotics	Osaka, Japan	Jan 2022 – Mar 2023	Built the bin-picking pose estimator in PyTorch.
Machine Learning Intern	Naniwa Data Lab	Osaka, Japan	Jul 2021 – Dec 2021	Six months part-time alongside the bridge role, which is how the transition was funded. Built the baseline segmentation model later open-sourced as the bridge crack dataset. Presented weekly to a reading group of nine engineers on papers from CVPR and MVA.
Structural Engineer	Kansai Bridge & Structure	Kobe, Japan	Apr 2014 – Dec 2021	Load assessment for 60 prefectural road bridges under the 2016 revision of the Specifications for Highway Bridges. Led the seismic retrofit design for the Muko River crossing, a 340m continuous steel box girder. Ran finite-element models in Abaqus for fatigue life on orthotropic decks; published the calibration set. Supervised four junior engineers and the drawing office through two prefectural audits. Wrote the Python tooling that started the career change: a crack-width measurement pipeline over inspection photos that replaced two days of manual tracing per bridge. Standardised the inspection photo archive, 240k images, which later became the training set at Rinku. Represented the firm on the prefectural committee for bridge asset management, 2019-2021.

Education

MEng Civil Engineering, Kyoto University Apr 2012 – Mar 2014
Certificate in Applied Statistics, Osaka Institute of Technology Apr 2021 – Dec 2021
BEng Civil Engineering, Kyoto University Apr 2008 – Mar 2012

Projects

Bridge crack segmentation pytorch, opencv
Open dataset and baseline; 1.1k GitHub stars.

Certifications

Professional Engineer, Civil (Japan) — JSCE, Mar 2018
Deep Learning Specialisation — Coursera, Sep 2021
JDLA Deep Learning for Engineers — JDLA, Jul 2022

Publications

Crack width estimation on orthotropic steel decks from inspection photography. Journal of Bridge Engineering, Jun 2021
Transfer learning from civil inspection imagery to industrial defect detection. MVA, Jul 2023
A calibration set for INT8 export of small segmentation models. workshop paper, Nov 2024